

# High-Cost Driver and Prevention from Claims and Survey

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Data Scientist Applicant

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# Introduction

- Yirang Liu
- Master in Data Science from Duke University
- Master and Bachelor in Mechanical Engineering
- 7-year full-time experience as Senior Product Engineer at Cummins
  - Premise Health was the onsite clinic
- Internships
  - Data
    - GSK Capstone Predictive
    - Lenovo Machine Learning Optimization Engineer
  - Engineering: Volkswagen, Baker Hughes, SinoMach (Mercedes), Spectrum Brands, Inteva Products

# Agenda

- Acknowledgement
- Background and Business Understanding
- EDA
- Predictive modeling
- Agentic AI
- Potential topics
- Closing

# Acknowledgement

Yana: Hiring Manager

Emily: HR

Clay      My Precious

Nathan busy and

Nick professional

Shae audience



# Background and Business Understanding

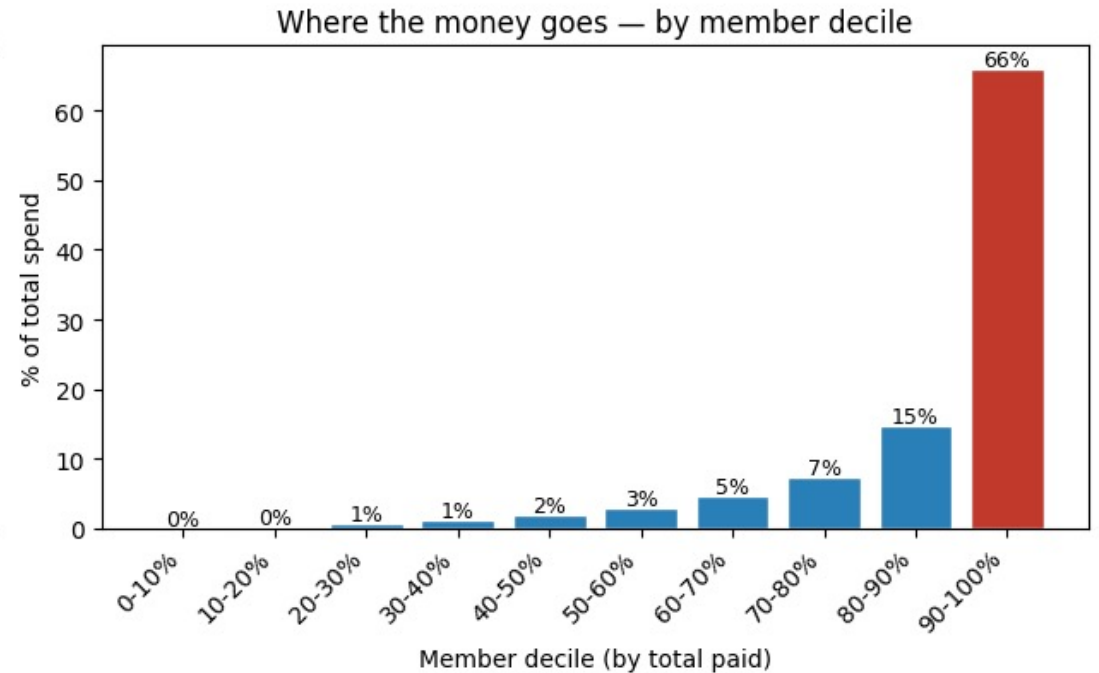
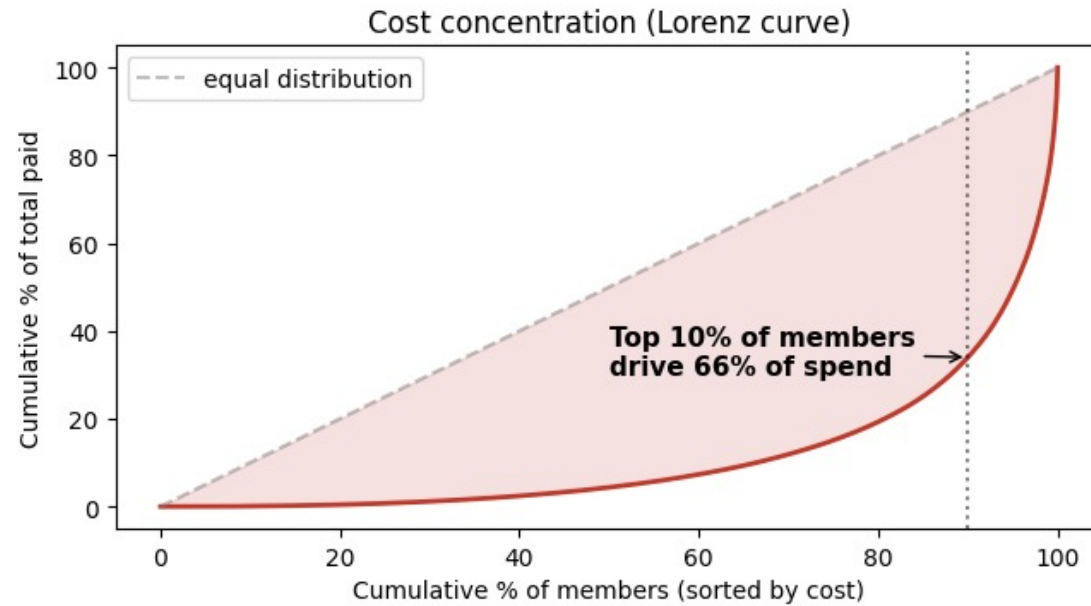
- Background: Interesting case study based on two datasets
  - Claims: 144247 rows
  - Survey: 4783 rows
- Project Statement
  - Understand the high-cost allocation and correlation to conditions
  - Help future budget allocation
  - Help prevent high cost and high acuity situation
- Scope and Limitation
  - Focusing on condition prevention to contribute better health in population and business profits

# Data Understanding

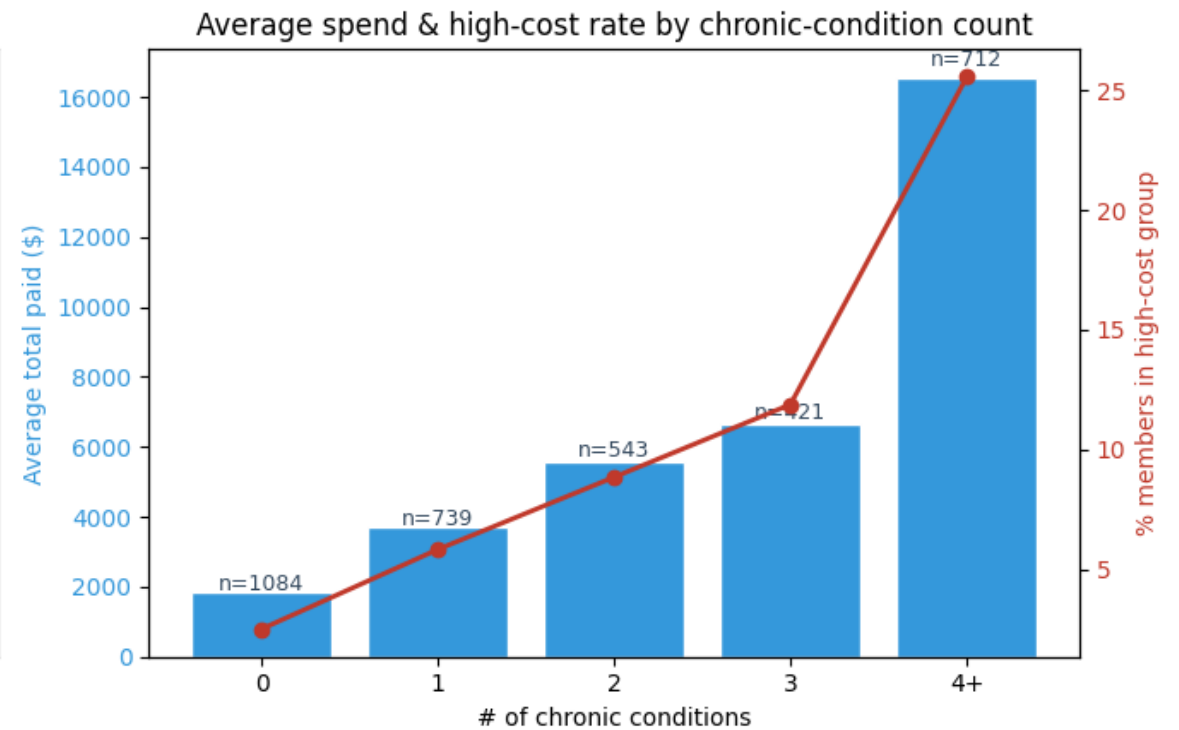
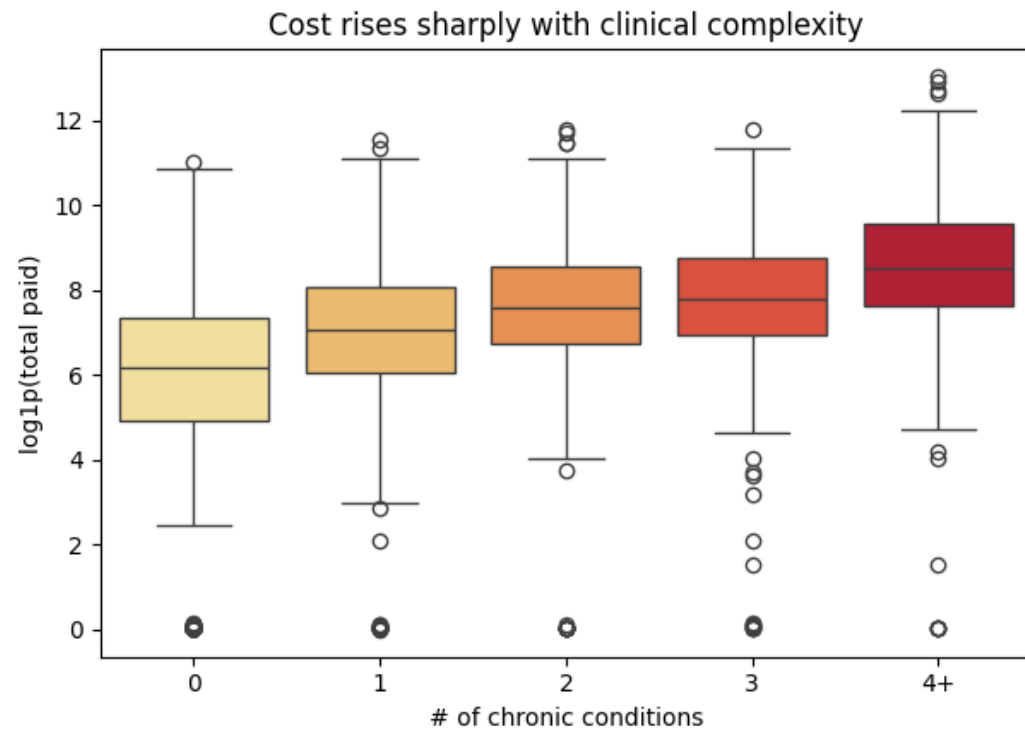
- Data Type Conversion
- Data Cleaning
  - Missing Values
  - Duplication
  - Outliers
- Data Exploration
  - Statistics
  - Balance
  - Potential Topics
- Takeaway:
  - Clean and imbalanced data as expected
  - Resources heavily inclined to
    - Top patience
    - Top condition
    - Top care
    - Top diagnosis
  - Not aligned with the focus of outpatient, primary clinic, predictive health

# Cost Concentration on Top Patients

Healthcare spend is highly concentrated in a few members

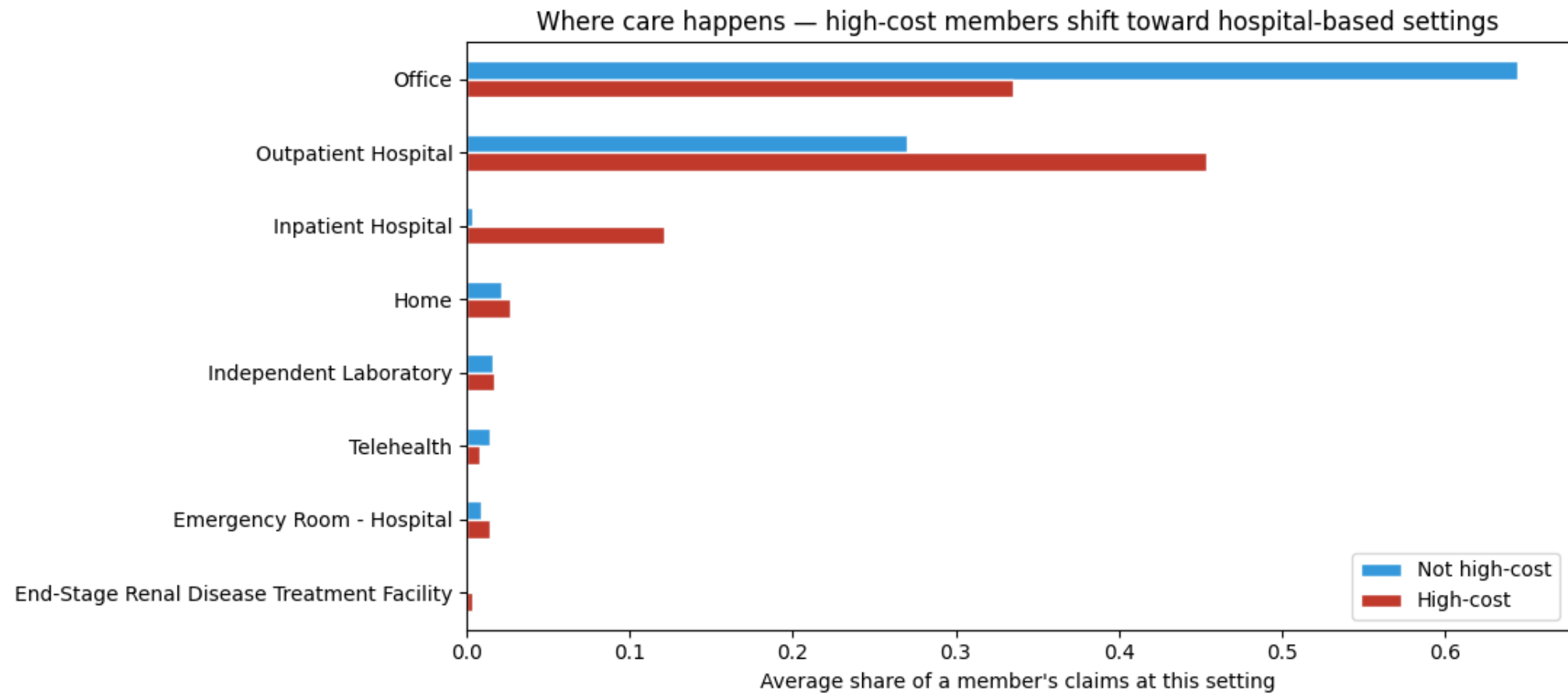


# Cost Concentration on Top Conditions

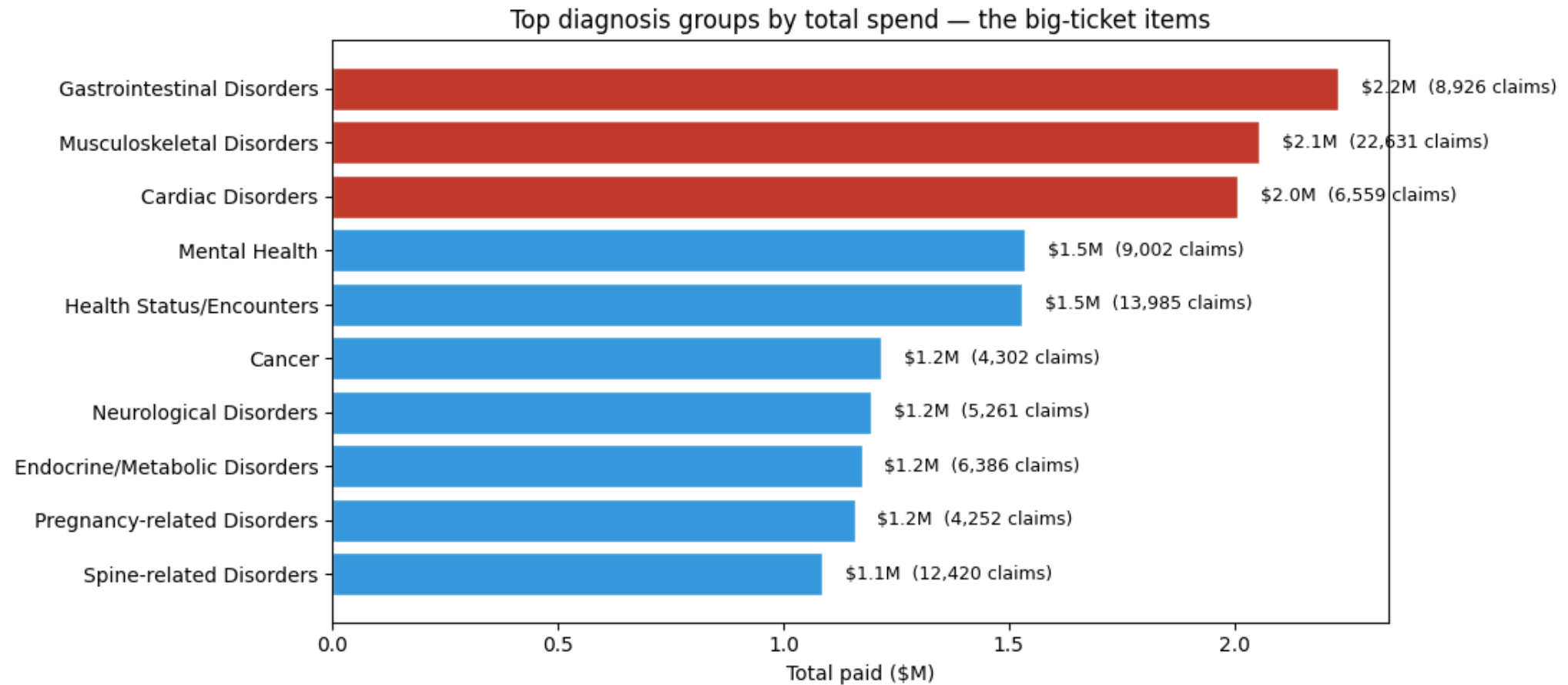




# Cost Allocation over Care



# Cost Allocation over Diagnosis



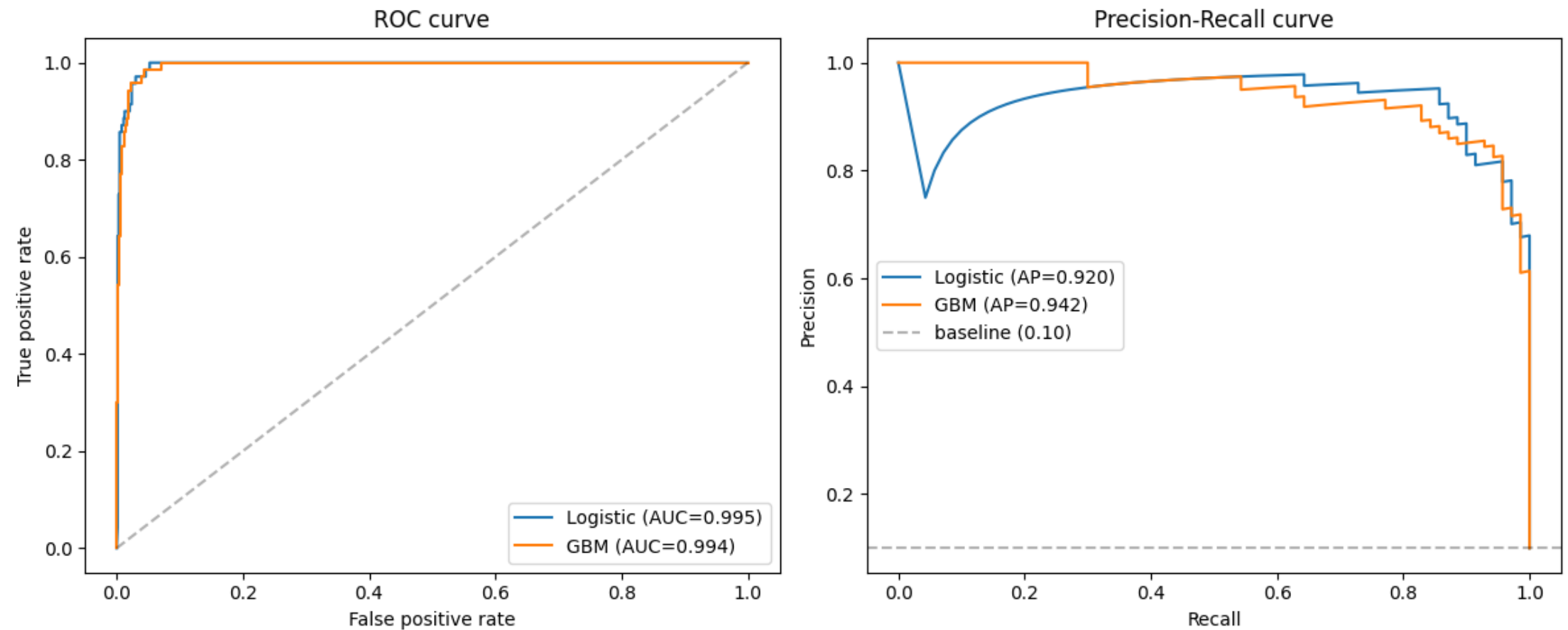
# Predictive modeling – High-Cost Member

- Drive
  - insurer and provider perspective
    - Budget and efficiency
  - Social responsibility
    - Better health overall
- Expectation
  - Find high-cost drivers
  - establish preventive action for healthy members

# Predictive modeling – High-Cost Member

- Data Split:
  - Stratified split (imbalance)
  - Train : test = 80 : 20
- Models:
  - Logistic Regression (baseline maybe underfitting)
  - Gradient Boosting (advance maybe overfitting)
- Metrics:
  - ROC-AUC + PR-AUC
  - ROC + PR curve plots

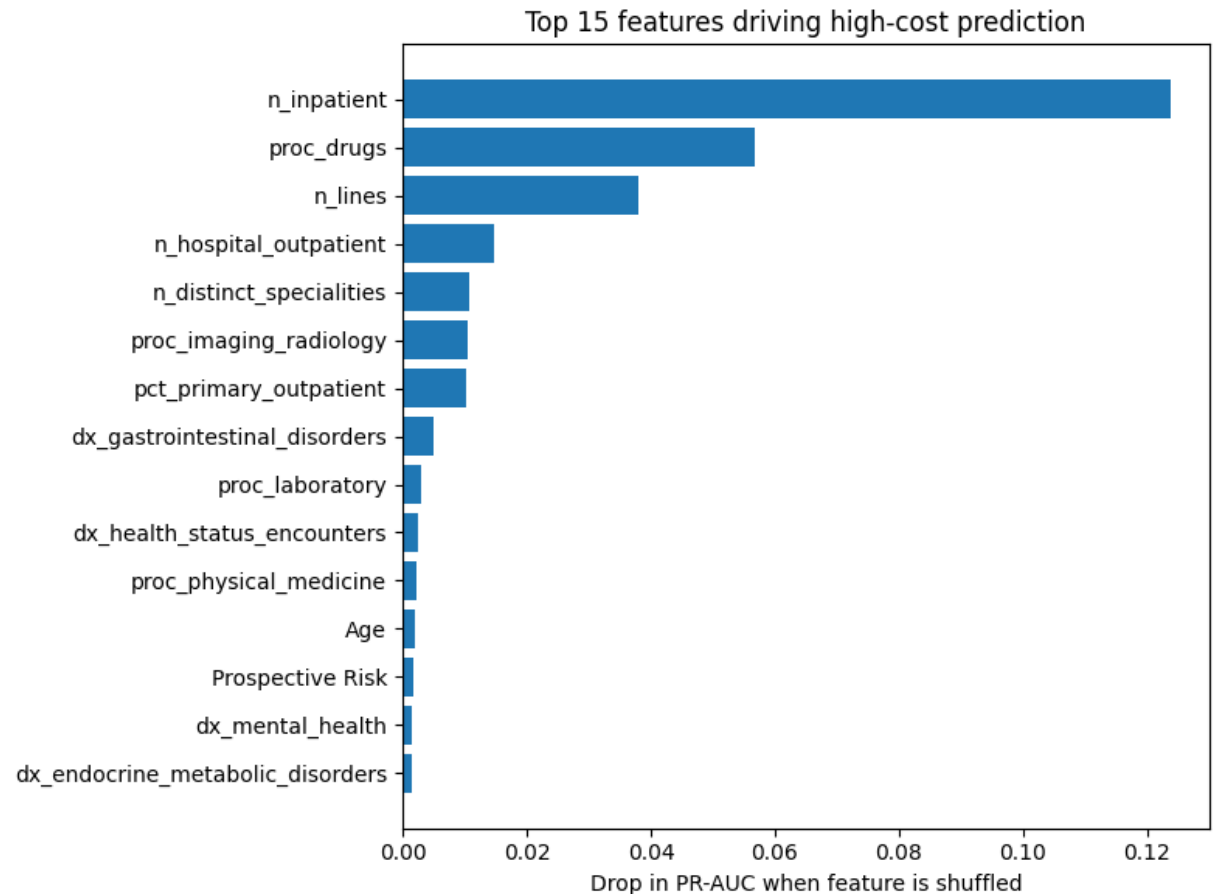
# Predictive modeling – Metrics



# Predictive Modeling – High-Cost Drivers

## Top 5 drivers

- Frequent visits
- Distinct specialties
- Occupation and behavior
- Mental Health
- Chronic leading to High Acuity



# Predictive Modeling – Takeaways

- Focus on primary care, outpatient, and telehealth
- Occupational and behavioral campaign
- Mandatory and incorporated mental health screening
- Disease management to prevent chronic leading to high acuity

# Future Work

- People who did not complete the survey
  - Customer satisfaction
- People did not have any claims
  - Completely healthy who don't need services but contribute to profit
  - Not sick enough to have to receive services contributing to profit
- People were not members
  - More promotion and marketing
  - Word of mouth from good services



Stay Well